AI and the Distribution of Prosperity:

A Unified Framework for Understanding How Technology Shapes Inequality*

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Abstract

How the gains from AI are distributed could have profound economic, social, and political consequences for decades to come. This paper develops a unified framework for understanding how AI affects income inequality. We begin by decomposing inequality into three components: labor-income concentration, capital-income concentration, and the capital-labor split. We then organize the mechanisms through which AI affects these components into three stages: (i) the *direct effects on production*, including automation, augmentation, new-task creation, changes in entry barriers, and capital deepening; (ii) the *distribution of productivity gains* among adopting firms, AI suppliers, workers, and consumers, and within each group; and (iii) the *economy-wide demand effects* generated by the spending and saving of AI's beneficiaries. Across these mechanisms, we identify the key factors that determine their direction and magnitude and synthesize the available evidence to date. Finally, we discuss how these mechanisms can combine to produce concentrated gains or shared prosperity, and identify key empirical indicators that can signal which direction the economy is heading.

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1 Introduction

Artificial intelligence has the potential to affect the economy in profoundly different ways. At one extreme, it could raise productivity and wages across the economy, make many essential services broadly accessible, and rebuild the middle class (Autor, 2024). At the other, it could concentrate enormous gains among a handful of superstar firms, wealthy asset owners, and highly paid workers, while leaving the majority of workers behind (Acemoglu, 2025). Where the economy ultimately falls between these two extremes will have profound economic, social, and political consequences for decades to come.

Recent history offers a cautionary example. Over four decades of computerization, the internet, and industrial robots, incomes at the top of the U.S. distribution grew rapidly, while incomes for the bottom half mostly stagnated or even declined (Piketty et al., 2018; Moll et al., 2022). As a result, the top 1% increased its share of national income from less than 12% in 1978 to around 20% in 2018, while the bottom 50% experienced the opposite trend (Saez and Zucman, 2019).[1]

Understanding and predicting AI's distributional consequences requires identifying both the mechanisms through which AI affects inequality and the factors that determine their direction and magnitude. Yet discussion of these mechanisms is often fragmented. Different mechanisms—such as machines replacing workers, AI making workers more productive, firms earning higher profits, goods getting cheaper, or the returns flowing to whoever owns the technology—are frequently discussed in isolation. At the same time, the sheer number of possible mechanisms and forces can make the question seem overwhelmingly complex.

This paper therefore develops a unified framework for thinking about how AI shapes the income distribution. Rather than asking simply whether AI will increase or decrease inequality, we ask two more nuanced questions: through which economic mechanisms can AI affect inequality, and which factors determine the direction and magnitude of their effects? Although the framework is developed around AI, it applies more broadly to technological change. We therefore draw throughout the paper on evidence from earlier technological advances.

[1] The magnitude of these patterns is debated: using different assumptions, Auten and Splinter (2024) estimate a smaller rise in the top 1% share. Nevertheless, under either estimate, pre-tax top income shares rose while the bottom half's share declined.

We begin in Section 2 by decomposing income inequality into three components: the concentration of labor income, the concentration of capital income, and the division of national income between labor and capital. We establish where each stands on the eve of the AI era: wage inequality is high by historical standards, the labor share has declined substantially over the past two decades, and capital income is extremely concentrated following decades of rising wealth concentration.[2]

Section 3 then organizes the main mechanisms through which AI can affect these components into three closely interconnected stages. The first stage (Section 3.1) consists of *direct effects on production*: automation, augmentation, new-task creation, changes in entry barriers, and capital deepening, closely paralleling the five categories of technological change proposed by Acemoglu et al. (2026). The distributional effects of these mechanisms depend primarily on AI's capabilities and on how workers and firms adopt and use it. The second stage (Section 3.2) concerns *the distribution of the productivity gains generated by AI adoption*: how they are distributed among AI suppliers, adopting firms, workers, and consumers—as well as within each of these groups—through higher profits, higher wages, and lower prices. This distribution depends largely on competition and bottlenecks along the AI supply chain, firms' market power, workers' bargaining power, and the responsiveness of consumer demand to prices. Finally, the third stage (Section 3.3) consists of *economy-wide demand effects*: as the initial beneficiaries of AI spend or save their gains, they reshape demand for different goods and services and, in turn, for the workers who produce them. These effects depend on who directly receives AI's gains and how those gains are spent or saved.

Together, these three stages trace how AI's economic effects unfold: how it changes production, how the resulting productivity gains are initially distributed, and how those gains subsequently reshape demand throughout the economy. At each stage, we identify the key technological and economic factors that determine the direction and magnitude of AI's distributional effects and synthesize the available evidence to date. Figure 1 sum-

[2]Throughout this paper, we focus on *pre-redistribution* income. We do so for two main reasons. First, technology shapes the income distribution primarily through markets—wages, profits, and prices—so its direct effects are reflected mostly in the pre-redistribution income distribution. Taxes and transfers are the policy response to those effects, and informing that response is one of this paper's objectives. Second, as Appendix A.1 shows, post-tax inequality has historically moved almost one-for-one with pre-tax inequality, both across countries and within them over time. In the United States, redistribution offset only a small fraction of the post-1980 rise in top income shares (Piketty et al., 2018).

marizes the framework.[3]

We then draw the framework together in Section 4 by tracing two illustrative scenarios: one in which the mechanisms combine to produce concentrated gains, and one in which they produce shared prosperity. These are not forecasts but stylized illustrations of how the same technology can generate very different distributional outcomes, depending on how the key technological and economic factors evolve. Finally, we identify key empirical indicators that can reveal which path the economy is taking, describing the data sources that already track them, and highlighting remaining measurement gaps.

The paper's central contribution is to integrate several literatures that have largely developed separately into a unified framework for assessing AI's overall distributional effects. Our analysis of AI's direct effects on production builds on the task-based literature on technology and labor markets (Acemoglu and Autor, 2011; Acemoglu and Restrepo, 2018; Moll et al., 2022; Autor and Thompson, 2025; Danieli, 2026; Acemoglu et al., 2026). Our analysis of how the resulting productivity gains are distributed draws on research on market power, rent-sharing, and pass-through (Melitz and Ottaviano, 2008; Card et al., 2018; Kline et al., 2019; De Loecker et al., 2020; Autor et al., 2020a), the economics of the AI supply chain (Korinek and Stiglitz, 2019; Korinek and Vipra, 2025), and the distributional effects of relative prices (Baumol, 1967; Jaravel, 2019). Finally, our analysis of economy-wide demand effects draws on macroeconomic evidence on saving, household debt, and the composition of consumption (Mian et al., 2021b; Buera et al., 2022; Moll et al., 2022; Hubmer, 2023; Fagereng et al., 2024). In each stage, we connect the effects to the three inequality components. This integration allows us to examine how mechanisms typically studied in isolation combine to determine AI's overall effect on the income distribution.

We hope this framework proves valuable for three main audiences. First, it can help organize the public discussion on AI and inequality by bringing together the main mechanisms through which AI may shape the distribution of income into a single coherent framework. Second, it provides policymakers with a structured way to think about AI's distributional consequences. By making the relevant mechanisms explicit, it clarifies which mechanisms different policies can influence and which indicators should be monitored as the technology diffuses. Third, it may help guide future research by identifying

[3] An initial version of an interactive companion to the framework, designed to illustrate its key mechanisms and their distributional implications, is available at `https://ai-distribution.org`. It allows users to explore alternative scenarios for the framework's mechanisms. The tool is under active development.

the empirical and theoretical questions that matter most for understanding how AI affects the distribution of income.

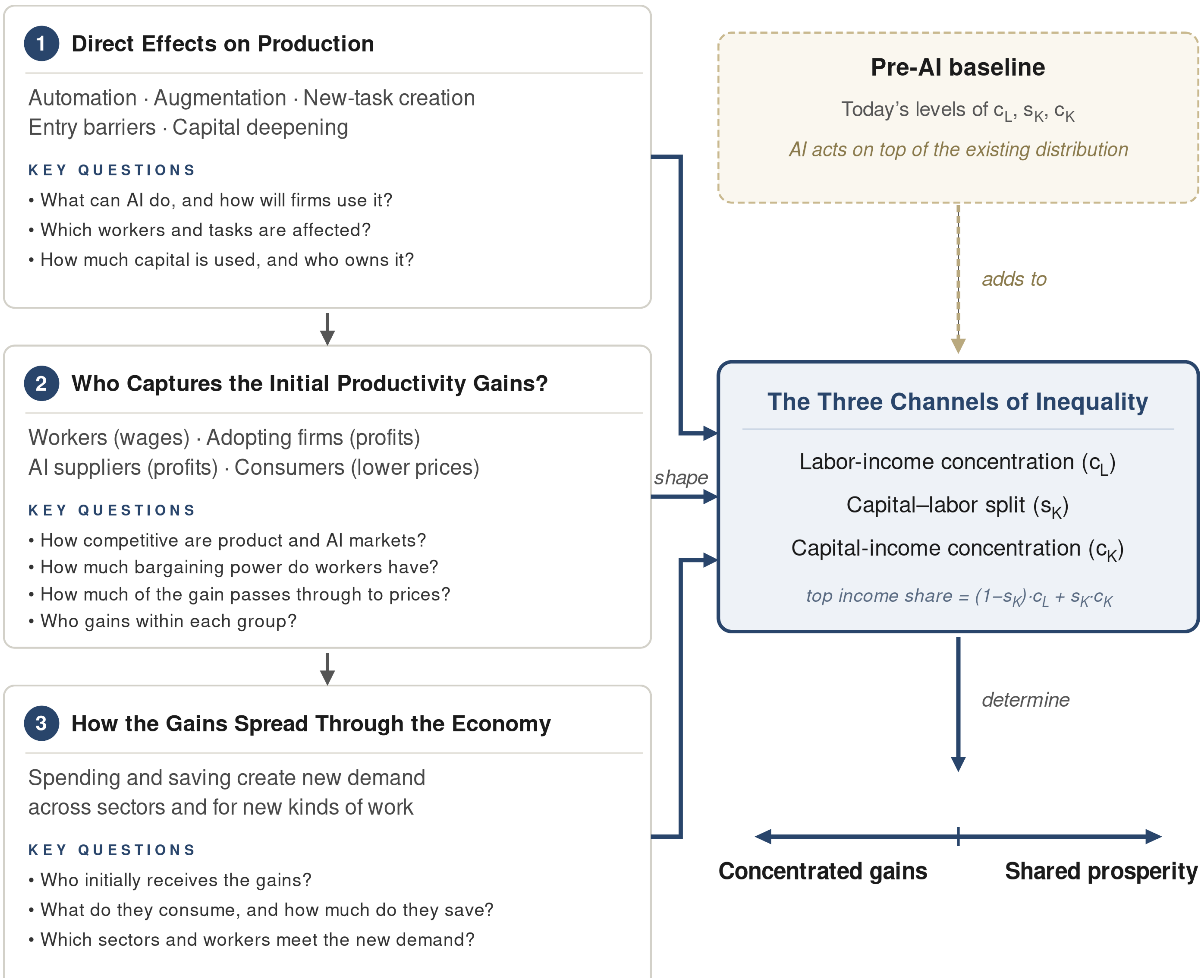

Figure 1: How AI Shapes the Distribution of Prosperity

AI's direct effects on production (1), the division of the resulting gains (2), and their spread through the economy (3) together shape the three channels of inequality—labor-income concentration (c_L), the capital–labor split (s_K), and capital-income concentration (c_K). Acting on top of the pre-AI distribution, these channels determine where outcomes fall between concentrated gains and shared prosperity.

2 The Three Channels of Income Inequality

Overview: Decomposing Income Inequality

Every person's income comes from two sources: labor income (wages) and capital income (returns on what they own).[4] The share of total income received by the top group (for example, the top 1%) can then be written as:

$$S_{\text{top}} \;=\; (1 - s_K)\, c_L \;+\; s_K\, c_K,$$

where s_K is the share of income paid to capital, $(1 - s_K)$ is the share paid to labor, and c_L and c_K are the top group's shares of all labor income and of all capital income (Meade, 1964; Moll et al., 2022).[5] In words, inequality (in this case, top income share) is a weighted average of *labor-income concentration* and *capital-income concentration*, weighted by each source's *income share*. An analogous decomposition applies to other inequality measures, such as the Gini coefficient (see Appendix A.3).

Decomposing changes over time in S_{top}, following Moll et al. (2022), gives three channels through which AI can act:

$$\Delta S_{\text{top}} \;=\; \underbrace{\Delta c_L\,(1 - s_K)}_{\text{wage concentration}} \;+\; \underbrace{\Delta c_K\, s_K}_{\text{capital concentration}} \;+\; \underbrace{\Delta s_K\,\big(c_K^{\text{post-AI}} - c_L^{\text{post-AI}}\big)}_{\text{shift toward capital}}$$

- **Labor-income concentration (Δc_L).** AI can change how unequally wages are spread across workers. This effect matters more the larger the labor share $(1 - s_K)$.
- **Capital-income concentration (Δc_K).** AI can change how concentrated capital income is across individuals. This effect matters more the larger the capital share (s_K).
- **Shift toward capital (Δs_K).** AI can change the division of income between labor and capital. This effect matters more the larger the gap between capital-income and labor-income concentration ($c_K^{\text{post-AI}} - c_L^{\text{post-AI}}$).[6]

[4] Capital income consists mainly of interest, dividends, rents, and business profits, including retained corporate profits attributed to shareholders.

[5] The algebraic derivation is provided in Appendix A.2.

[6] Note that $c_K^{\text{post-AI}} = c_K + \Delta c_K$ and $c_L^{\text{post-AI}} = c_L + \Delta c_L$ are the post-AI concentrations.

The remainder of this section examines each of the three channels in the decomposition in turn: labor-income concentration, capital-income concentration, and the capital–labor split.[7] For each, it explains what it measures, why it matters for inequality, and how it evolved in the decades preceding AI. This establishes the pre-AI baseline against which the AI-driven changes analyzed in Section 3 can be evaluated.

2.1 Distribution of Labor Income Across Workers (c_L)

c_L is the concentration of labor income across workers—the share of total wage income accruing to a top income group. Wages are the main source of income for most households: outside the top decile, labor income far exceeds capital income, so how wages are distributed largely determines living standards for most people. Earnings also shape economic security, social status, and economic mobility.

Recent trends: U.S. wage inequality rose sharply after 1980. The college wage premium widened substantially, but inequality also increased among workers with similar education and experience. The rise was especially pronounced at the top: the gap between high earners and the middle widened much more than the gap between the middle and the bottom (Katz and Murphy, 1992; Autor et al., 2008, 2020b; Katz, 2025).

Drivers: Technological change was one of the main drivers of rising wage inequality. Computers complemented workers performing analytical and managerial tasks while replacing many routine clerical and production tasks, contributing to the decline of middle-paying jobs (Autor et al., 2003; Acemoglu and Autor, 2011). One estimate attributes 50–70% of the rise in wage inequality since 1980 to the automation of routine work (Acemoglu and Restrepo, 2022). Wage losses were greatest in middle-income routine occupations where technology replaced workers' core skills and expertise (Autor and Thompson, 2025; Danieli, 2026). Moreover, the new, high-paying tasks created by technology became increasingly concentrated in well-paid professional and technical jobs requiring higher education (Autor et al., 2024, 2026). Weaker labor-market institutions reinforced these effects, particularly at the bottom of the distribution: the declining real value of the minimum wage reduced pay among less-educated workers, while falling unioniza-

[7] Note that AI can also affect the distribution of real purchasing power through changes in relative prices, as discussed in Section 3.2.2.

tion weakened workers' bargaining power (Cengiz et al., 2019; Farber et al., 2021; Vogel, 2025). Trade also played an important role. Import competition from China (the "China shock") caused sharp and persistent declines in manufacturing employment and earnings (Autor et al., 2013, 2016, 2021).

Looking ahead: Wage inequality therefore enters the AI era high by historical standards. Past experience suggests that AI's effects will depend not only on what the technology can do, but also on labor-market institutions and broader macroeconomic conditions.

2.2 Distribution of Capital Income Across Individuals (c_K)

c_K is the concentration of capital income across individuals: the share of profits, interest, rents, dividends, business income, and other returns to wealth received by those at the top. Capital income is considerably more concentrated than labor income, making it especially important for inequality at the top, and its importance grows as the capital share rises (Section 2.3). The wealth underlying capital income also provides economic security, political influence, and advantages that can be passed across generations.

Recent trends: Capital income has become an increasingly important source of inequality. Since 2000, most of the rise in the top 1% income share has come from capital income, particularly retained corporate profits (Piketty et al., 2018). Over the same period, wealth remained extremely concentrated and became substantially more so. Estimates suggest that the top 0.1% owned roughly 16–22% of U.S. wealth by the early 2010s—up from about 7% in the late 1970s—while the share owned by the bottom 90% fell from approximately 35% to 23% (Saez and Zucman, 2016; Smith et al., 2023).

Drivers: Capital income depends on both the amount of wealth households own and the returns they earn on it, both of which are highly concentrated at the top. First, wealth ownership is highly unequal because wealthy households save a larger share of their income and therefore accumulate wealth more rapidly (Mian et al., 2021b; Straub, 2019). Second, wealthy households earn systematically higher returns on their assets (Fagereng et al., 2020; Smith et al., 2023), in part because their portfolios are more heavily concentrated in business equity, causing equity-price gains to accrue disproportionately to them (Kuhn et al., 2020). As a result, capital income is even more concentrated than wealth itself (Gaillard et al., 2023). Automation further reinforces this pattern by raising demand for capital, thereby increasing returns to wealth (Moll et al., 2022).

Looking ahead: Capital income therefore enters the AI era already highly concentrated. Whether AI further increases this concentration will depend on who owns AI-related capital, how its returns are distributed, and whether opportunities to own that capital become broadly accessible.

2.3 Distribution of Income Between Capital and Labor (s_K)

s_K is the share of national income paid to owners of capital. The remainder, the labor share $(1 - s_K)$, is paid to workers through wages and other compensation. Because capital income is much more concentrated than labor income, a larger capital share generally shifts more income toward those at the top of the distribution (Moll et al., 2022).

Recent trends: Once considered one of the stable features of economic growth, the labor share has fallen substantially in recent decades. In the United States, it remained near 63% through much of the late twentieth century before declining by roughly 6 to 8 percentage points and reaching its lowest level of the postwar period (Grossman and Oberfield, 2022; Audoly et al., 2026). Similar declines occurred across many other countries (Karabarbounis and Neiman, 2014).

Drivers: The causes of this decline remain debated (Grossman and Oberfield, 2022). One explanation is technological change: as capital equipment became cheaper, firms substituted capital for labor by shifting tasks performed by workers to machines (Karabarbounis and Neiman, 2014; Hubmer, 2023; Acemoglu and Restrepo, 2022; Moll et al., 2022). Other explanations emphasize changes in market structure: highly productive "superstar" firms with relatively low labor shares gained market share (Autor et al., 2020a), while rising markups increased the share of output captured as profits, at the expense of labor (De Loecker et al., 2020; Barkai, 2020). Together, these forces have widened the gap between productivity and pay across advanced economies (Suss, 2026).

Looking ahead: The labor share therefore enters the AI era after decades of decline. AI could push it lower if it substitutes for workers or increases profits and economic rents, but it could raise the labor share if it strongly complements workers and increases their wages. Because capital income is highly concentrated, a further shift toward capital would generally increase inequality (Piketty, 2014; Autor and Kausik, 2026).

3 How AI Shapes the Income Distribution

The three channels introduced in Section 2—the concentration of labor income (c_L), the concentration of capital income (c_K), and the share of income paid to capital (s_K)—provide the framework for the rest of the paper. This section examines the economic mechanisms through which AI may change these channels and the factors that determine the direction and magnitude of its effects.

We organize the mechanisms into three interconnected stages. The first, *direct effects on production* (Section 3.1), examines how AI changes what work is performed, who performs it, and how capital-intensive the production is. It includes automation, augmentation, new-task creation, changes in occupational entry barriers, and capital deepening, closely paralleling the five categories of technological change identified by Acemoglu et al. (2026). The second stage, *how productivity gains are distributed* (Section 3.2), examines how the gains generated by AI are divided among workers, adopting firms, AI suppliers, and consumers through higher wages, higher profits, and lower prices. The third stage, *economy-wide demand effects* (Section 3.3), examines how the spending and saving of those who initially benefit from AI reshape demand for goods and services, and therefore for the workers who produce them. For each mechanism, we set out what it is and the key questions that determine its effect on inequality, and we synthesize the available evidence to date.

No single mechanism determines AI's overall effect on the income distribution; instead, the eventual outcome depends on how these mechanisms interact. As Section 4 illustrates, different combinations of these mechanisms can produce very different outcomes, ranging from concentrated gains among a narrow set of firms, owners, and top earners to broadly shared prosperity through higher wages and lower prices. Keeping these two broad outcomes in mind provides a useful lens for the discussion that follows: depending on the underlying technological and economic conditions, each mechanism can contribute to either one. Because the mechanisms are closely interconnected, the discussion includes occasional overlap and cross-references across the following subsections.

3.1 Direct Effects on Production

Table 1 previews the five mechanisms through which AI directly reshapes production. For each, it provides a brief description and the key distributional questions it raises, which organize the discussion that follows. We then examine each mechanism in detail and the main factors that determine its effect on inequality.

Table 1: Summary of the Direct Effects on Production

Mechanism	Description	The key distributional questions we discuss
Automation (§3.1.1)	AI performs existing tasks end-to-end with little human intervention, substituting capital for labor.	(i) How far does automation extend? (s_K) (ii) Which workers are directly affected? (c_L) (iii) Where do displaced workers move? (c_L) (iv) Who owns the capital that gains? (c_K) (v) Who captures the productivity gains? (c_L, s_K, c_K)
Augmentation (§3.1.2)	AI raises the productivity of workers in tasks they continue to perform.	(i) Which workers experience the largest productivity gains, within and across occupations? (c_L) (ii) Who captures the productivity gains? (c_L, s_K, c_K)
New-task creation (§3.1.3)	AI creates entirely new tasks and occupations.	(i) How much new work is created? (s_K) (ii) Who gets the new work? (c_L, s_K, c_K)
Changes in entry barriers (§3.1.4)	AI changes the expertise needed to perform tasks or enter occupations, altering occupational wage premiums.	(i) Where are barriers lowered or raised? (c_L) (ii) Who can access and use AI effectively? (c_L) (iii) Do institutions allow boundaries to change? (c_L)
Capital deepening (§3.1.5)	AI increases the capital used per worker through AI adoption and capital-intensive AI production.	(i) How much does AI increase capital intensity? (s_K) (ii) Does AI substitute for or complement labor? (s_K) (iii) Who owns the capital and captures its returns? (c_K)

Notes: Each mechanism is examined in the subsection referenced beside its name, together with the main factors that determine its effect; the questions correspond one-to-one to the correspondingly numbered parts of that subsection. The symbols in parentheses indicate the inequality channel(s) each question bears on most directly: the concentration of labor income (c_L), the concentration of capital income (c_K), and the share of income paid to capital (s_K), as defined in Section 2.

3.1.1 Automation (Labor Substitution)

Automation is technological change that allows capital—machines, software, and now AI—to perform tasks previously carried out by workers. In the AI context, automation occurs when AI substitutes for human labor by performing a task end-to-end with little human intervention. By contrast, when AI assists with a task that still depends substantially on human judgment or intervention, it augments the worker rather than automating the task (see Section 3.1.2). Automation is therefore the most direct way AI can reduce demand for labor in affected tasks, shift income from labor toward capital, and change which workers and skills are valued (Acemoglu and Restrepo, 2018; Moll et al., 2022; Korinek and Stiglitz, 2019).

AI automation affects inequality through three linked effects (Acemoglu and Restrepo, 2018, 2022). The first is a *displacement effect*: capital takes over tasks previously performed by workers, shifting those tasks from labor to capital. The second is a *ripple effect*: displaced workers move into other occupations, increasing labor supply there and potentially lowering wages even for workers whose own tasks were not automated. The third is a *productivity effect*: if AI performs automated tasks more effectively than workers, it can increase firms' productivity, allowing them to produce more output with the same inputs or the same output at lower cost.

The distributional consequences of these three effects depend primarily on five questions: (i) how far automation extends; (ii) which workers are directly affected; (iii) where displaced workers move; (iv) who owns the capital that benefits; and (v) who ultimately captures the resulting productivity gains.

(i) How Far Automation Goes

The extent of automation is a central determinant of its direct effect on the capital–labor split (s_K): the more tasks firms shift from workers to AI, the more income is likely to move from labor toward capital. One task-level assessment classifies roughly 20% of U.S. tasks as highly automatable, with many more at least partly exposed.[8]

Technical exposure, however, does not necessarily translate into automation. Very few tasks can currently be performed end-to-end without human oversight (Eloundou et al., 2024), and end-to-end automation remains a small share of actual use (Iscenko et al., 2026). Whether firms automate an exposed task depends on AI's capabilities and reliability, the cost of compute relative to wages, and the costs of reorganizing production

[8]These figures are based on LLM-generated task-level ratings of Hosseini and Lichtinger (2026a), who update the widely-used O*NET tasks scored by Eloundou et al. (2024) for automation potential on a five-level scale (T0–T4); see Appendix A.4 for details. They should be interpreted with caution as a measure of technical exposure rather than a forecast of actual displacement.

around the technology (Lashkari et al., 2026). Looking further ahead, AI may also be used to improve AI and robotics themselves, expanding the range of tasks that can be automated over time. How far automation ultimately extends will be a key determinant of how much income shifts from labor toward capital.

(ii) Which Workers Are Affected

The effect of AI automation on wage inequality (c_L) depends largely on where the most exposed occupations lie in the wage distribution. Current estimates suggest that automation exposure is concentrated in information-processing and clerical occupations, including insurance claims, data entry, proofreading, and travel services. Occupations centered on physical dexterity, in-person care, or hands-on judgment appear less exposed to current AI systems (Hosseini and Lichtinger, 2026a; Eloundou et al., 2024).

Figure 2 shows how the estimated share of tasks that can be automated within an occupation varies across the wage distribution. Which occupations are most exposed depends heavily on how far AI capabilities advance. A limited wave, covering only tasks AI can already perform end-to-end, would have relatively small distributional effects. A modest wave, covering tasks for which current AI can perform most components, would primarily affect the clerical middle, resembling earlier patterns of job polarization.[9] A broader wave of more capable AI would increasingly reach high-paying professional and managerial work. Physical and in-person occupations remain the least directly exposed under all three scenarios, although they could still be affected through ripple effects.

However, the share of tasks automated is only part of the story. The distributional effects also depend on which tasks within an occupation AI automates. Automating an occupation's core expert tasks may erode its wage premium, whereas automating routine supporting tasks may allow workers to focus on higher-value activities (Autor and Thompson, 2025; see Section 3.1.4). Within occupations, workers whose skills better complement the remaining non-automated tasks are likely to benefit more from AI (Freund and Mann, 2026).

[9] The automatable share rises sharply around the 35th wage percentile (roughly $43,000 annually) because customer service representatives and general office clerks are among the largest and most automatable clerical occupations, together employing roughly five million workers.

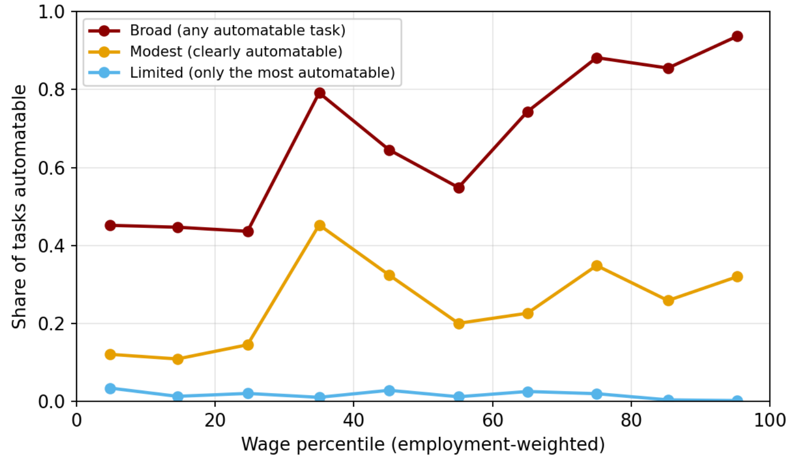

Figure 2: Estimated Share of Automatable Tasks by Occupational Wage Decile

Notes: The figure shows the employment-weighted average share of tasks within occupations estimated to be automatable by AI. These estimates are based on LLM-generated task classifications and should be interpreted as illustrative. The limited scenario includes only tasks classified as completely automatable; the modest scenario includes tasks classified as mostly automatable; and the broad scenario includes all tasks classified as at least partly automatable. Data are from Hosseini and Lichtinger (2026a), building on Eloundou et al. (2024). See Appendix A.4 for details on the data and methodology.

(iii) Where Displaced Workers Move

The distributional consequences of automation depend not only on which workers are affected, but also on where they move afterward. Workers with more transferable skills are generally better positioned to adapt to displacement (Manning and Aguirre, 2026). Highly paid professionals such as software developers, financial analysts, and managers often possess skills that transfer readily across occupations, making them better positioned to adjust. By contrast, workers in exposed lower-middle-paying occupations—including customer service representatives, general office clerks, secretaries, and receptionists—typically have fewer outside options, making wage losses or unemployment more likely following displacement.

The effects also extend to workers whose own tasks were never automated through *ripple effects*, which depend on where displaced workers seek new employment. For example, if workers displaced from lower-middle-paying clerical occupations move into lower-paying service jobs, the resulting increase in labor supply may depress wages in those occupations. Automation can therefore reduce wages even for workers whose own tasks were never automated, as occurred during earlier waves of technological change (Acemoglu and Restrepo, 2020, 2022).

(iv) Who Owns the Capital That Gains

If automation shifts income from labor to capital, its effect on capital-income concentration (c_K) depends on who owns the capital that benefits (Moll et al., 2022). AI-related capital is likely to be owned either by firms adopting AI or by firms supplying models, chips, compute, and other infrastructure. To the extent that ownership of these firms remains concentrated among wealthy households, automation will increase capital-income concentration. Automation also increases demand for capital and thereby may raise returns to wealth—especially risky assets—more broadly, disproportionately benefiting wealthy households that own it (Kuhn et al., 2020; Moll et al., 2022).

(v) Who Captures the Productivity Gains

Finally, automation can generate productivity gains by allowing firms to produce more efficiently and at lower cost. As discussed in detail in Section 3.2, the distributional implications of these gains depend on who captures those gains: workers through higher wages, adopting firms through higher profits, AI suppliers through the prices they charge, or consumers through lower prices. Gains concentrated in corporate profits are more likely to increase inequality because ownership is highly concentrated, whereas wage gains and lower prices may benefit a broader population—although this depends on which workers receive higher wages and which consumers buy the affected goods. This division is determined largely by market conditions, including market power, bottlenecks, and the price elasticity of demand (Section 3.2).

3.1.2 Augmentation (Labor Complementarity)

Augmentation is technological change that makes workers more productive at tasks they continue to perform. For many tasks, AI may prove better suited to complementing workers than replacing them outright because it remains too unreliable, or too expensive, to perform the entire task end-to-end. In these cases, the worker remains responsible for completing the task, while AI provides assistance. Unlike automation, augmentation therefore does not directly displace labor from the task.

The distributional effects of AI augmentation depend primarily on two questions. The first is *which workers experience the largest productivity gains*: AI may generate different gains for workers with different levels of skill and experience within the same occupation, and its effects may also vary across occupations. However, higher worker productivity does not necessarily translate into higher wages. The second question is therefore *how the productivity gains are divided*.

(i) Which Workers Experience the Largest Productivity Gains

Within occupations, a growing body of experimental evidence suggests that AI often raises the productivity of less-experienced and lower-performing workers the most. Studies of customer support, professional writing, product development, and business problem-solving consistently find much larger gains for these workers than for their more experienced peers, substantially narrowing performance gaps (Brynjolfsson et al., 2025b; Noy and Zhang, 2023; Dell'Acqua et al., 2025; Cruces et al., 2026). These findings suggest that AI could act as an equalizer by extending expert-level performance to less specialized workers (Autor, 2024). However, most of the evidence comes from controlled experiments on narrow, self-contained tasks. In broader workplace settings, domain expertise and judgment may remain important determinants of who benefits most (Hitzig et al., 2026). Moreover, as AI becomes agentic—performing more of the work while humans provide direction and exercise judgment—its gains may increasingly favor experienced workers with more managerial skills (Alekseeva et al., 2026; Weidmann et al., 2025). Finally, augmentation may prove temporary: workers who benefit most from AI today may later become exposed to automation as AI capabilities improve (Manning, 2024).

Between occupations, task-based estimates suggest that the potential gains from AI augmentation rise with wages: they are largest in high-paying cognitive occupations and smallest in lower-paying physical and in-person work (Eloundou et al., 2024; Hosseini and Lichtinger, 2026a). Unequal adoption may reinforce this pattern: surveys show that AI use at work is roughly twice as common among college-educated workers as among workers without a college degree (Bick et al., 2024), and provider usage data show that actual workplace AI use concentrates in higher-paying cognitive occupations (Anthropic, 2025; Iscenko et al., 2026). As a result, even if AI compresses productivity differences within occupations, it may widen them across occupations by raising productivity most where earnings are already highest.

(ii) Who Captures the Productivity Gains

Higher productivity does not necessarily translate into higher wages or employment for augmented workers. Whether workers benefit depends partly on the *price elasticity of demand* for their output—how strongly demand responds when prices fall (Autor, 2015; Bessen, 2019). If demand expands substantially, firms may increase production enough to raise demand for their labor. If demand responds little, firms may instead meet existing demand with fewer workers, reducing labor demand. Tax preparation provides a simple example: because the number of tax returns is relatively fixed, AI assistance may primarily make filing cheaper and faster rather than increasing employment or wages among tax preparers.[10]

More broadly, as with automation, productivity gains from augmentation may accrue to workers through higher wages, adopting firms through higher profits, AI suppliers through higher prices, or consumers through lower prices. This allocation depends on market forces such as competition, workers' bargaining power, supply-chain bottlenecks, and the responsiveness of demand to lower prices. The distributional effects of AI augmentation therefore depend on how the productivity gains are allocated among and within these groups (see Section 3.2 for a detailed discussion).

3.1.3 New-task Creation

New-task creation is technological change that creates entirely new tasks and occupations. This is different from simply increasing demand for existing work: technology may generate more employment in existing occupations without creating genuinely new tasks, a mechanism discussed in Section 3.3. New-task creation matters because it is the main force counteracting automation. Whereas automation shifts existing tasks from workers to capital, new tasks can create new roles for workers and restore labor's share of income

[10] A classic historical example is agriculture: U.S. agricultural productivity rose dramatically during the twentieth century, but because demand for food is inelastic—people do not eat much more as food becomes cheaper—output did not expand enough to offset labor-saving productivity growth. Farm employment therefore fell even as agricultural output increased (Autor, 2015; Bessen, 2019).

(Acemoglu and Restrepo, 2018). Historically, new work has also commanded substantial wage premiums, making access to it an important determinant of wage inequality (Autor et al., 2026).

Because new tasks typically emerge only as technologies diffuse and firms reorganize production around them, it remains difficult to predict what new work AI will create or how important it will become. Nevertheless, its distributional consequences mainly depend on two questions: (i) how much new work AI creates, and (ii) who gets that new work. The first determines whether new-task creation is large and fast enough to offset automation. The second determines whether the gains flow to workers or capital owners and, if they flow to workers, which workers benefit.

(i) How Much New Work AI Creates

A central question is whether AI creates enough new work, and quickly enough, to offset the jobs it automates. Historically, technological change eventually generated substantial new work: more than 60% of U.S. employment in 2018 was in occupations whose job titles did not exist in 1940 (Autor et al., 2024). Some AI-related work is already emerging, including building and evaluating models, managing AI adoption within firms, creating and reviewing training data, and teaching others to use AI tools (Jaumotte et al., 2026; Adrjan, 2026). Whether this work becomes a major source of employment or remains relatively specialized alongside widespread automation remains uncertain.

(ii) Who Gets the New Work

The second question is who gets the new work. New tasks do not necessarily create jobs for people: AI systems may perform them from the outset, or soon after they emerge, allowing production to expand with little additional demand for labor and directing the gains toward capital owners (Acemoglu and Restrepo, 2018; Restrepo, 2025). When workers do perform the new tasks, the effect on wage inequality depends on who gets those jobs and how well they pay. Historically, new work has disproportionately employed younger and more-educated workers and commanded significant wage premiums, especially when associated with new technologies; since 1980, it has become increasingly concentrated in professional and technical occupations (Autor et al., 2026; Acemoglu et al., 2026). AI could therefore widen wage inequality if new tasks remain concentrated among highly educated specialists, or narrow it if they become accessible to a broader range of workers.

3.1.4 Changes in Entry Barriers and Expertise Requirements

One important determinant of an occupation's wage is how scarce the expertise required to perform it is (Autor and Thompson, 2025). Specialist doctors earn high wages not only because there is strong demand for their services, but also because there is a limited supply of workers who possess the necessary expertise, and acquiring it requires extensive training and talent. Many other well-paid occupations similarly command wage premiums because their expertise is scarce and hard to acquire. AI could reduce these premiums by making expert capabilities more widely available, allowing workers with less training to perform work previously reserved for specialists (Autor, 2024). However, AI could also raise entry barriers if using it effectively requires new technical, managerial, or coordination skills in occupations that did not previously require them.

The distributional effects depend primarily on three questions: (i) where AI lowers or raises expertise requirements; (ii) who can access and use AI effectively; and (iii) whether institutions allow occupational boundaries to change. Together, these determine who gains access to well-paid work and how large the resulting expertise premiums remain.

(i) Where AI Changes Expertise Requirements

AI could change expertise barriers in two ways. First, it may allow workers in one occupation to perform tasks previously reserved for another: with AI assistance, paralegals might perform some tasks currently done by lawyers, while nurses might perform some tasks traditionally performed by doctors (Autor, 2024). Second, it may change the expertise required to enter an occupation, allowing more or fewer workers to qualify for it. In both cases, changing the supply of qualified workers would affect the occupation's wage premium.

The effect on wage inequality (c_L) depends on where these changes occur. AI could lower expertise requirements in high-paying occupations and thereby narrow wage gaps by enabling more workers to perform high-value work (Hosseini and Lichtinger, 2026a; Althoff and Reichardt, 2026). By contrast, if AI raises the technical or managerial skills required to obtain entry-level or middle-paying jobs, it could place good jobs further out of reach and widen inequality. Early evidence points in this direction, particularly for entry-level jobs (Hill, 2026; Westby et al., 2026).

(ii) Access to and Adoption of AI

Lower expertise requirements will have little effect if access, training, or adoption costs prevent less-skilled workers from using AI effectively. Early evidence already suggests unequal adoption: workplace use of generative AI is roughly twice as high among workers with bachelor's or graduate degrees as among those with less education (Bick et al., 2024). Moreover, schools serving more economically disadvantaged students also have lower levels of AI integration, including less teacher training, guidance, and access to AI-enabled tools (Campos and Singleton, 2026). Such gaps could limit who benefits from lower expertise barriers.

(iii) Whether Institutions Allow Occupational Boundaries to Change

Technical feasibility alone does not determine who performs a job. Licensing, credential requirements, regulation, and professional gatekeeping may preserve occupational boundaries even when AI reduces the expertise needed to cross them. Scope-of-practice rules, for example, may prevent nurses from performing tasks traditionally reserved for physicians, while legal regulations may prevent workers without formal legal credentials from offering certain services. Employers and consumers may also continue to prefer traditionally credentialed professionals, even when less-credentialed workers can perform the work effectively with AI. Whether these institutions and preferences adapt alongside the technology will therefore determine whether lower expertise requirements translate into broader access to well-paid work (Autor, 2024; Acemoglu et al., 2026).

3.1.5 Capital Deepening

Capital deepening means that production uses more capital, or capital services, per worker. In the AI context, this capital includes compute, chips, data centers, software, frontier models, and cloud computing. Whereas automation and augmentation describe how AI affects workers' tasks, capital deepening describes how it affects the amount of capital used relative to labor.

AI can deepen capital in two ways. First, firms adopting AI may use more compute, software, and other AI-related capital for each worker they employ. An accounting firm that keeps the same number of accountants but equips each of them with AI software supported by cloud computing, for example, now uses more capital per worker. Second, producing AI is itself highly capital-intensive, requiring large investments in chips, data centers, and infrastructure (Goldman Sachs Research, 2025; International Energy Agency, 2025; Cottier et al., 2024). As the AI-producing sector grows, the economy may become more capital-intensive.

How capital deepening affects inequality depends primarily on three questions: (i) how much AI increases capital intensity; (ii) whether AI capital substitutes for or complements labor; and (iii) who owns that capital and captures its returns.

(i) How Much AI Increases Capital Intensity

The scale of capital deepening depends on both channels described above. Through AI adoption, it depends on how widely and intensively firms use AI-related capital and on the cost of compute, models, and software relative to labor. Through AI production, it depends on how capital-intensive the AI-producing sector remains and how large that sector becomes relative to the rest of the economy. Lower AI costs, broader adoption, and faster growth of AI production would increase the use of capital per worker, making the remaining distributional questions more consequential.

(ii) Whether AI Capital Substitutes for or Complements Labor

More capital per worker does not necessarily mean that a larger share of income goes to capital. If it complements labor strongly enough, it can raise workers' productivity and wages, and labor's share may increase rather than fall. However, if AI capital substitutes for labor—for example, by automating tasks previously performed by workers—it can reduce labor's share of income and raise the capital share (s_K). Studies of earlier waves of capital deepening reach different conclusions about whether substitution or complementarity historically dominated (Karabarbounis and Neiman, 2014; Hubmer, 2023; Lawrence, 2015).

Whether AI capital will be more substitutable for or complementary to labor than earlier technologies remains uncertain. Much AI development is aimed at replicating tasks currently performed by people, potentially making AI capital especially substitutable for labor (Brynjolfsson, 2022; Acemoglu et al., 2026). However, current AI systems often remain too unreliable to perform tasks end-to-end without substantial human verification and judgment, potentially making them better suited to complement workers than to replace them outright (Autor, 2024; Rabanser et al., 2026; Burn-Murdoch and O'Connor, 2026; Iscenko et al., 2026). Which force dominates—and therefore how AI affects the labor share—will depend on both AI's capabilities and how firms choose to deploy it.

(iii) Who Owns the Capital and Captures Its Returns

If capital deepening raises the capital share, its effect on inequality depends on who ultimately receives the additional capital income. The returns to AI-related capital may flow to firms adopting AI or to firms supplying chips, compute, models, and infrastructure. Because ownership of businesses and financial assets is highly concentrated, these returns are likely to accrue disproportionately to wealthy households, increasing capital-income concentration (c_K) (Piketty, 2014; Saez and Zucman, 2016; Moll et al., 2022). As discussed in Section 3.2.1, this effect is likely to be larger when bottlenecks and market power in the AI supply chain allow a few firms to capture a large share of the gains.

3.2 Who Captures the Productivity Gains?

As discussed in the previous section, AI can generate substantial productivity gains for adopting firms, for example by automating or augmenting workers' tasks.[11] This section examines how those gains are divided and the market forces that shape their allocation. To organize the discussion, we trace the gains downstream in two stages. First, Section 3.2.1 examines how AI suppliers, such as providers of models, compute, chips, and other key inputs, may capture part of the gross gains through the prices they charge, reducing the amount that remains for adopting firms. Second, Section 3.2.2 examines how the gains remaining in adopting firms after payments to AI suppliers are distributed

[11]By a productivity gain, we mean that AI allows a firm to produce more output with the same inputs, the same output with fewer inputs, or some combination of the two.

among firm owners, workers, and consumers—and within each group—through higher profits, higher wages, and lower prices. The allocation in both stages depends less on AI's technical capabilities than on market forces such as competition, workers' bargaining power, supply-chain bottlenecks, and the responsiveness of demand to lower prices. Figure 3 summarizes this two-stage process and the market forces governing each stage.

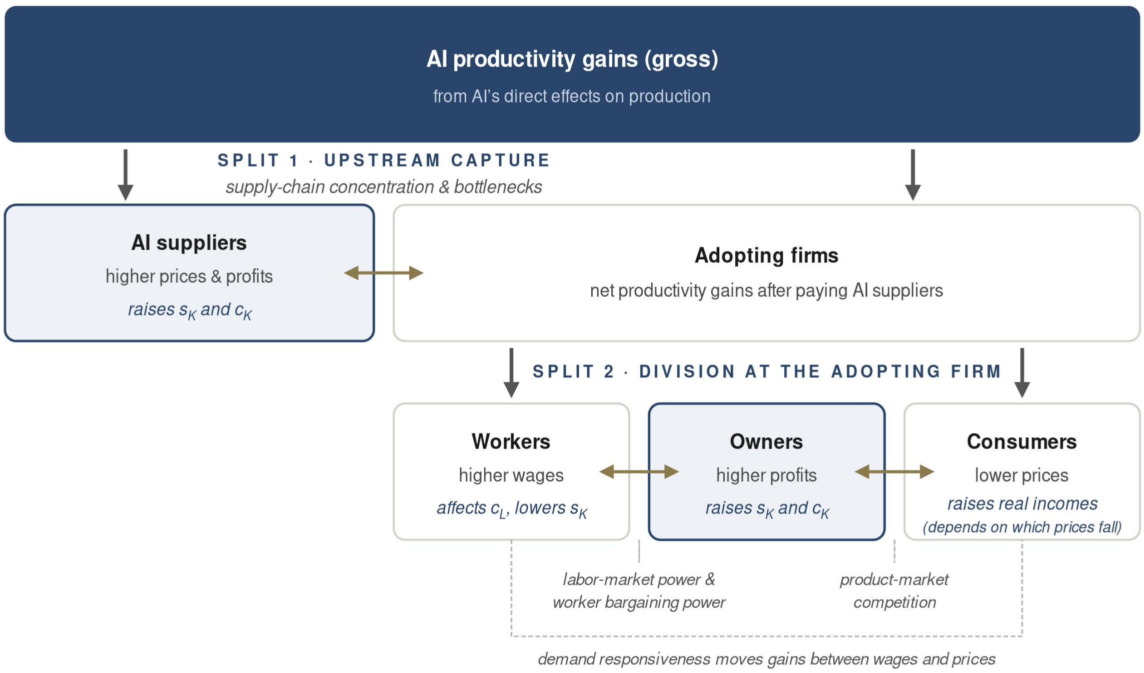

Figure 3: How AI's Productivity Gains are Divided

Notes: AI suppliers first capture part of the gross gains through the prices they charge (Split 1). The remaining gains are then divided at the adopting firm among workers, owners, and consumers (Split 2). Double-headed arrows mark the boundaries that each market force shifts. Box widths are illustrative and do not represent estimated shares.

3.2.1 Upstream Capture in the AI Supply Chain

When a firm adopts AI to raise productivity, it pays for models and other AI services supplied by companies such as OpenAI or Anthropic. The price it pays reflects both costs incurred throughout the AI supply chain—including chips, memory, and data centers—and any markups or scarcity rents captured by suppliers (Korinek and Stiglitz, 2019; Korinek and Vipra, 2025). This price is a key determinant of where AI's productivity gains accrue:

the more suppliers charge, the larger the share they capture at the expense of firms that adopt AI. We focus on two factors that shape how much suppliers can charge: (i) market power in the AI supply chain and (ii) bottlenecks in the supply of key inputs.

(i) Market Power in the AI Supply Chain

The AI supply chain is highly concentrated, with individual firms dominant at several key stages. The most advanced AI chips are designed largely by Nvidia, fabricated by TSMC using lithography equipment supplied primarily by ASML, and paired with specialized memory supplied by three firms (Sastry et al., 2024). Such concentration gives suppliers substantial pricing power. Nvidia's gross margin, for example, has recently been around 75% (NVIDIA Corporation, 2026). When suppliers can sustain margins of this magnitude, they capture a larger share of AI's productivity gains as profits, leaving less for adopting firms, workers, and consumers.

(ii) Bottlenecks in the AI Supply Chain

Another factor determining how much AI suppliers can charge is the extent of bottlenecks along the supply chain. A bottleneck arises when demand for an input grows faster than the industry's ability to produce it, pushing its price up sharply even when no single firm dominates the market. This dynamic is already visible throughout the AI supply chain. The main constraints are not the logic chips themselves but the advanced packaging (TSMC's CoWoS) and high-bandwidth memory needed to build them, both of which remain supply-constrained, and increasingly electricity generation and grid capacity (Somala, 2026; International Energy Agency, 2025). These inputs scale slowly: new fabrication and packaging lines, power plants, and grid connections take years to build, while demand for AI compute has been growing rapidly (Somala, 2026). These constraints can allow upstream suppliers to capture part of the gains from cheaper and more capable AI models through higher input prices and scarcity rents, rather than allowing those gains to pass fully to adopters.

Distributional Implications

Whether driven by market power or bottlenecks, upstream capture has important implications for inequality. Most of the gains are likely to accrue as profits, increasing the capital share (s_K). Because corporate equity is disproportionately owned by wealthy households, these profits are also likely to increase capital-income concentration (c_K). Some gains also accrue to workers employed by AI suppliers. But because these workers are among the highest-paid in the economy (Semiconductor Industry Association, 2021; NVIDIA Corporation, 2025; Bhaimiya, 2025), these gains are likely to increase the concentration of labor income (c_L). How important these channels prove in the long run will depend on how competition evolves (Korinek and Vipra, 2025)—for instance, whether cloud providers' in-house chips or new entrants erode Nvidia's position—and on the extent to which today's bottlenecks are eventually eased.

3.2.2 How Adopting Firms Divide the Gains

After paying AI suppliers for their services, adopting firms are still likely to retain substantial productivity gains.[12] These gains can then be shared with workers through higher wages, passed on to consumers through lower prices, or retained by owners as higher profits. How these gains are divided turns on three margins, each shaped primarily by a different market force: (i) the split between wages and profits, set by the balance of power in the labor market; (ii) the split between prices and profits, set by competition in the product market; and (iii) the split between wages and prices, set by how strongly consumer demand responds to lower prices. Put differently, the allocation of productivity gains is a question of pass-through: how much firms pass on to workers through higher wages and to consumers through lower prices, rather than retain as profit.[13]

(i) Labor-Market Power: the Profit–Wage Split

Whether an adopting firm's AI gains reach its workers or stay with owners depends

[12] Note that AI could also affect firms that never adopt: adopters may gain market share at rivals' expense, while downstream firms could benefit from cheaper inputs. These spillovers operate through the same margins—wages, profits, and prices—discussed in this section.

[13] Our discussion follows models of competition with variable markups in the tradition of Melitz and Ottaviano (2008). In these models, the prices firms can charge are constrained by the alternatives available to their customers. The fewer and weaker a firm's competitors, the smaller the share of a cost reduction it must pass on through lower prices (for empirical evidence, see Amiti et al., 2019). The same logic applies in the labor market: the wage a firm must pay is disciplined by workers' outside options, so the fewer employers competing for its workers, the less of a productivity gain it must hand over in higher wages (Berger et al., 2022).

on the balance of labor-market power between them. When workers are hard to replace, can easily move to rivals, and have good outside options, the firm must share the gains to keep them; but where the labor market is concentrated, or search frictions are significant, the firm can pay workers below the value they produce and keep the gains as profit (Caldwell et al., 2026; Kline, 2025). Unions can further strengthen workers' leverage by pressing firms to distribute more of the gains through higher wages (Jäger et al., 2025). The long decline of unionization in the United States has therefore left workers less well positioned to capture the gains from AI (Farber et al., 2021). Moreover, AI may itself shift the balance of power further toward employers: AI-enabled monitoring, algorithmic management, and predictive wage-setting can reduce workers' bargaining leverage and allow firms to retain a larger share of the gains (Davis, 2024).

(ii) Product-Market Power: the Price–Profit Split

Whether the gain reaches consumers as lower prices depends on competition in the product market. Where competition is strong, a larger share of cost savings is passed through into lower prices; where firms have market power, they can retain more of the savings as profit. Over recent decades, product-market markups have risen and are considered one explanation for the decline in the labor share (De Loecker et al., 2020). Whether AI reinforces this tilt toward profits or reverses it will depend on how competitive the sectors experiencing its biggest productivity gains prove to be— and on how AI itself reshapes competition through its effects on entry and innovation.

(iii) Product-Demand Responsiveness: the Wage–Price Split

Beyond market power, the effect of productivity gains on wages and prices depends on how consumer demand for the firm's output responds. If demand is elastic, productivity gains lead to a large expansion of output, so firms demand more labor, and workers can gain through higher employment and pay. However, if demand is inelastic, output expands little, so firms need fewer workers to produce the same level of output, and labor demand—and therefore employment and wages—falls (see the detailed discussion in Section 3.1.2; Autor, 2015; Bessen, 2019). In this case, more gains pass through into lower prices or higher profits, with the division between the two determined by the price–profit split described above. Demand responsiveness thus shapes a further split, between workers and consumers: whether higher productivity translates into higher wages or into lower prices.

Distributional Implications

The distributional consequences of AI-driven productivity gains at adopting firms depend both on how these gains are divided among firm owners, workers, and consumers and on how they are distributed within each group.

Gains that are retained by firm owners as profits, whether because of labor-market power or market power in product markets, increase the capital share (s_K). Because corporate ownership is highly concentrated, they are also likely to increase capital-income concentration (c_K). The magnitude of these effects depends not only on how much profit is generated but also on *which firms* capture it. If AI's gains accrue disproportionately to a few dominant firms, the capital share can rise through reallocation as economic activity shifts toward large, high-markup firms with relatively low labor shares: the "superstar firm" mechanism (Autor et al., 2020a). Likewise, the more the gains accrue to firms with concentrated ownership—for example, closely held or founder-dominated firms rather than widely held corporations—the more they increase capital-income concentration.

Gains that reach workers instead increase the labor share. Their effect on wage inequality (c_L), however, depends on which workers receive them. Workers with greater bargaining power, and those employed in sectors where demand is more elastic, are likely to capture a larger share of the gains. High earners often have greater bargaining power because they possess harder-to-replace skills and better outside options (Kline et al., 2019). Whether high- or low-paid workers are concentrated in sectors with more elastic demand remains an open empirical question.

Gains passed on through lower prices increase real incomes, and because lower-income households spend a larger share of their income on consumption, a broad price decline benefits them proportionally more. But the distributional implications depend on *which* prices fall and who consumes those goods. AI may reduce prices primarily in the cognitive and digital sectors it can reach, while labor-intensive services such as healthcare, childcare, and education—where productivity grows slowly—could become relatively more expensive, reflecting "Baumol's cost disease" (Baumol, 1967).[14] If AI-exposed goods

[14]This pattern is already visible across rich economies: prices of tradable goods have fallen sharply while

consumed disproportionately by higher-income households become cheaper, while essential services remain expensive, AI's price effects could widen real-income inequality—unless AI eventually reaches in-person services themselves.[15]

3.3 Economy-Wide Demand Effects

The previous sections examined how AI reshapes production and who captures the productivity gains. But those gains do not stop where they first accrue. Higher wages and profits raise household income, while lower prices raise households' purchasing power. Households then spend or save these gains. Spending creates demand for goods and services, and therefore for the workers who produce them, while saving accumulates as financial assets and may finance investment or borrowing elsewhere. As a result, a productivity gain in one part of the economy can raise labor demand in another, so the ultimate winners and losers need not be the workers whose tasks AI directly affects. This demand channel has shaped previous technological transitions: as productivity growth made households richer, they spent a rising share of their income on services such as healthcare, restaurant meals, and financial services, and employment shifted from farms and factories toward those services (Autor, 2015; Buera et al., 2022; Hubmer, 2023).

The distributional consequences of these economy-wide effects depend on three questions: (i) who receives the initial gains; (ii) whether they spend or save them; and (iii) what they spend them on, and which sectors—and workers—meet the demand they create. Figure 4 summarizes this chain.

(i) Who Receives the Gains?

The demand channel starts where Sections 3.1 and 3.2 ended: with the division of AI's productivity gains among workers, owners, AI suppliers, and consumers, and within each group. What matters here is which households ultimately receive those gains. Wage gains accrue to particular groups of workers, profits accrue disproportionately to wealthy shareholders, and lower prices benefit the households that consume the affected goods.

those of essential services have soared, raising the share of middle-class income spent on essentials from roughly one-third to one-half (Burn-Murdoch, 2025).

[15] The price benefits of product innovation in recent decades accrued disproportionately to high-income households, generating sizable inflation inequality (Jaravel, 2019, 2021; Chakrabarti et al., 2026).

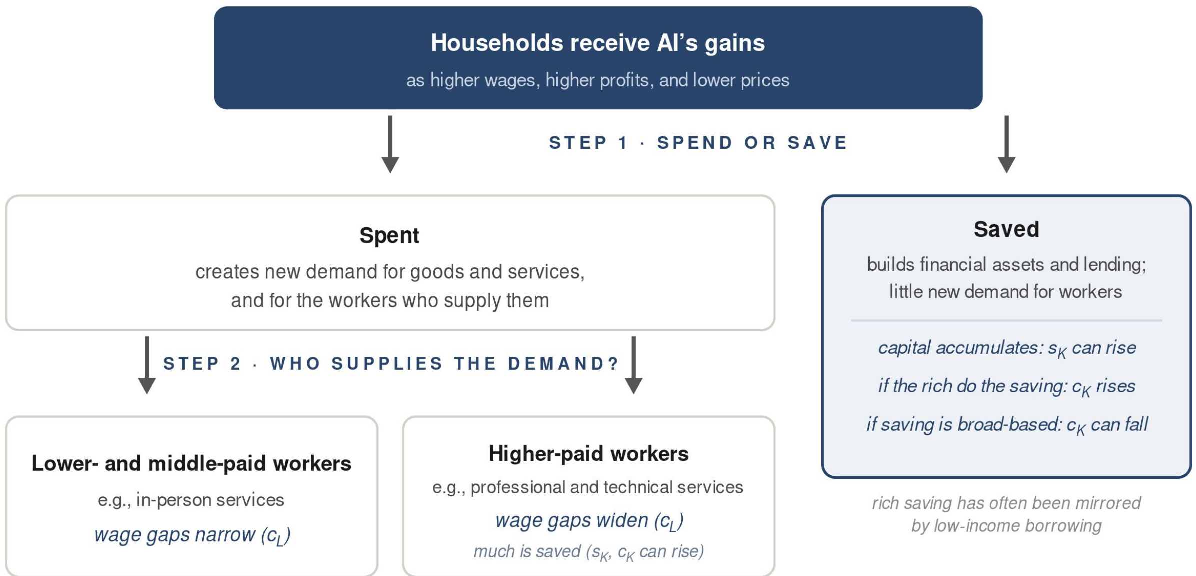

Figure 4: How AI's Gains Translate into New Demand

Notes: The figure follows AI's productivity gains after their initial division among workers, owners, and consumers (Section 3.2). Households spend or save the additional income (Step 1). Spending creates new demand for goods and services, and for the workers who supply them; whether it narrows or widens wage inequality (c_L) depends on whose labor meets the new demand (Step 2). Saving generates little demand for workers and instead builds financial assets: as capital accumulates, the capital share (s_K) can rise, while the effect on capital-income concentration (c_K) depends on who does the saving. Historically, high saving by wealthy households has been accompanied by greater borrowing among lower-income households (Mian et al., 2021b).

This mapping determines whose income and purchasing power rise, and therefore whose spending and saving decisions shape the new demand.

(ii) Do They Spend or Save?

The second question is how much of these gains are spent rather than saved. Lower- and middle-income households spend a much larger fraction of an additional dollar of income than high-income households, who save much of it (Dynan et al., 2004; Straub, 2019; Mian et al., 2021b). Consequently, if AI's gains accrue primarily to high-income households and capital owners, they generate less demand for consumption and labor than if the same gains were distributed more broadly (Auclert and Rognlie, 2018; Mian et al., 2021a). Nor do higher savings by high-income households necessarily finance new productive capital: over recent decades, much of the rise in top-end saving was absorbed by household and government borrowing rather than productive investment (Mian et al., 2021b). Moreover, much of this additional saving may instead bid up the prices of existing assets, reinforcing wealth concentration at the top (Auclert et al., 2025; Fagereng et al., 2024; Gomez and Gouin-Bonenfant, 2024; Kuhn et al., 2020).

(iii) What Do They Spend Their Income On, and Who Meets That Demand?

The third question concerns the composition of the additional spending. Richer households devote larger shares of their budgets to labor-intensive services—from restaurant meals and travel to education—and smaller shares to necessities such as utilities and food consumed at home (Hubmer, 2023). At the same time, their spending tilts toward skill-intensive services—healthcare, education, finance, and business services, whose share of economic activity rises systematically with income (Buera et al., 2022).

These spending patterns determine which sectors expand and, in turn, which workers experience stronger labor demand. If AI's gains accrue primarily to high-income households, the resulting demand is likely to favor skill-intensive services and the highly educated workers they employ. By contrast, if spending expands mainly in labor-intensive, in-person services such as care, hospitality, and food service, it is more likely to support employment and wage growth among low- and middle-income workers.

Distributional Implications

The distributional consequences of AI's economy-wide demand effects depend on who receives the initial gains, how much they spend rather than save, and where their spending is directed.

When gains accrue disproportionately to high-income households, a larger share is likely to be saved. These savings can reinforce wealth accumulation at the top and may finance greater borrowing by low- and middle-income households, potentially increasing capital-income concentration (c_K) (Mian et al., 2021b). By contrast, when AI's gains accrue more broadly to low- and middle-income households, a larger share is likely to be spent, generating stronger demand for goods, services, and labor. The saving that does occur is spread across a broader range of households, which may broaden asset ownership and reduce capital-income concentration over time.

For the portion that is spent, the distributional implications depend on its composition. If additional spending shifts primarily toward services, it is likely to support the labor share $(1 - s_K)$, since services are relatively labor-intensive (Hubmer, 2023). Which services expand also matters: if demand rises mainly for skill-intensive services—such as healthcare, education, finance, and legal and business services—it is likely to increase demand for highly educated workers and widen labor-income inequality (c_L) (Buera et al., 2022). By contrast, spending directed toward in-person and middle-wage services may spread labor-market gains more broadly across low- and middle-income workers.

4 Synthesis: Concentration or Shared Prosperity

Section 3 examined the main mechanisms through which AI may shape income inequality. Its overall distributional effect will depend on how those mechanisms combine. This section brings the framework together in two steps. Section 4.1 traces the two poles of the spectrum: one in which the mechanisms reinforce concentrated gains (Section 4.1.1), and another in which they reduce inequality and generate shared prosperity (Section 4.1.2). These are not forecasts but illustrative scenarios, and the actual outcome will almost certainly combine elements of both. They nevertheless illustrate how the same technology can produce very different distributional outcomes and clarify the forces that will determine which path the economy follows. Figure 5 summarizes the two scenarios. Section 4.2 then asks how we can tell which world is emerging by identifying the indicators worth monitoring, the data available to track them, and the most important remaining measurement gaps.

4.1 Two Polar Scenarios

4.1.1 A World of Concentrated Gains

In the first world, the direct effects on production (Section 3.1) largely favor the top. For high-paid professionals, AI automates the routine parts of the job and augments the core: it drafts, summarizes, and analyzes while the expert directs and judges, raising their productivity and the value of their expertise. For many middle-paying occupations, by con-

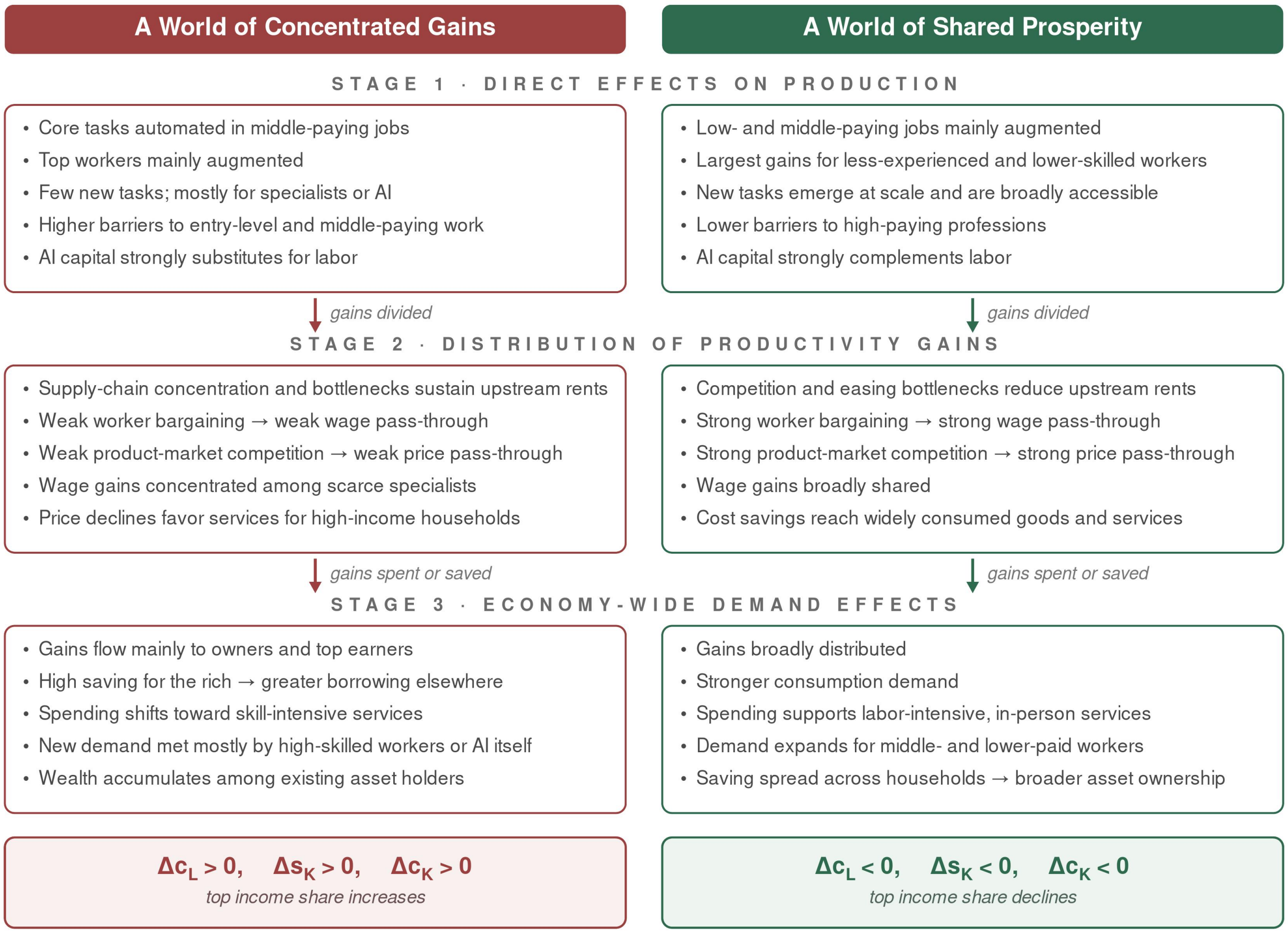

Figure 5: Summary of the Two Polar Scenarios

Notes: Each column summarizes one scenario through the three stages of the framework. See Section 4.1 for details. These are not forecasts but illustrative scenarios; the actual outcome will almost certainly combine elements of both. Their goal is to illustrate how AI can produce very different distributional outcomes and clarify the key forces that will determine which path the economy follows.

trast, AI automates the very tasks that justified their wage premiums. New tasks emerge only slowly, and those that do are performed mainly by highly skilled workers or by AI itself. At the same time, using AI effectively requires new technical and managerial skills, making entry into well-paid occupations more difficult rather than easier. Production also becomes increasingly capital-intensive as firms substitute compute for labor.

In this scenario, a large share of the resulting productivity gains (Section 3.2) is captured within the AI supply chain. Persistent bottlenecks in chips, compute, and energy—together with the concentration of the firms that control them—allow AI suppliers to charge high prices and capture substantial economic rents. Adopting firms with market power retain much of what remains as profit, while workers with weak bargaining power receive relatively little. The wage gains that do occur accrue mainly to the scarce specialists who build, manage, and direct AI systems. At the same time, AI lowers prices primarily in cognitive and digital services—such as legal and financial services—which are consumed disproportionately by higher-income households. In contrast, labor-intensive services such as childcare, healthcare, and education become relatively more expensive because productivity grows more slowly in those sectors (Baumol's cost disease).

The gains therefore accrue disproportionately to shareholders and high-income workers, who save a large share of the additional income (Section 3.3). This reinforces wealth accumulation at the top and may finance greater borrowing by low- and middle-income households. The spending that does occur is concentrated in skill-intensive services, further increasing demand for highly skilled workers. Some of the new demand is also met by AI itself, generating additional returns for AI suppliers and capital owners.

In this world, all three channels reinforce one another: wage inequality rises ($\Delta c_L > 0$), income shifts from labor to capital ($\Delta s_K > 0$), and capital income becomes even more concentrated ($\Delta c_K > 0$). Because capital income is already much more concentrated than labor income ($c_K > c_L$), every term in the decomposition of Section 2 pushes in the same direction, amplifying income concentration. Importantly, this outcome does not require a break with history. It extends the trends documented in Section 2: rising wage inequality, a declining labor share, higher markups, and growing wealth concentration. Concentrated gains are therefore not a departure from the recent past, but an acceleration of it.

4.1.2 A World of Shared Prosperity

In the second world, AI's direct effects on production (Section 3.1) broadly favor workers, especially those in the middle and lower parts of the wage distribution. Reliability constraints—or deliberate deployment choices—limit end-to-end automation, and AI is used primarily to augment workers rather than replace them. It takes over routine tasks while helping workers perform the judgment-intensive parts of their jobs, with the largest productivity gains accruing to less-experienced and lower-skilled workers. AI also lowers barriers to scarce expertise, enabling less-trained workers to perform tasks previously reserved for specialists and reducing wage premiums in high-paying professions. Together, these effects narrow productivity and wage gaps both within and between occupations. At the same time, new tasks emerge at scale and are accessible to a broad range of workers rather than concentrated among a small group of highly educated specialists.

In this scenario, the resulting productivity gains (Section 3.2) are broadly shared. Competition among AI providers, falling model prices, open alternatives, and easing compute bottlenecks erode upstream rents. Competitive product markets pass much of the remaining gains on as lower prices, including for the goods and services that dominate ordinary households' budgets. At the same time, workers throughout the wage distribution have enough bargaining power to capture a meaningful share through higher wages.

Because these gains accrue primarily to low- and middle-income households—who tend to spend a larger share of their income on consumption—a larger share is spent rather than saved (Section 3.3). This spending supports demand for goods and services and for the workers who provide them, particularly in labor-intensive, in-person sectors—such as care, hospitality, and parts of education—where AI remains less able to substitute for labor and many lower- and middle-paid workers are employed. The income that is saved, meanwhile, is spread across a broad range of households, allowing more families to accumulate financial assets, broaden capital ownership, and finance productive investment.

In this world, all three channels reduce inequality: wage inequality narrows ($\Delta c_L < 0$), the labor share rises ($\Delta s_K < 0$), and capital income becomes less concentrated ($\Delta c_K < 0$). The top income share therefore declines. Unlike the concentrated-gains scenario, this outcome would require a reversal of several long-running trends: the automation of middle-income work, weakening competition, declining worker bargaining power, increasingly concentrated capital ownership, and unequal access to technology-driven productivity gains. Shared prosperity is possible, but it would require a substantial departure from the economic trajectory of recent decades.

4.2 What to Watch: Indicators of the Direction

Identifying which distributional path is emerging has important policy implications. This section examines the indicators that can reveal which direction the economy is heading. Section 4.2.1 organizes them according to the three stages of the framework, production, the division of productivity gains, and economy-wide demand effects, and the three inequality channels (see Table 2 for a summary). Section 4.2.2 then reviews the main data sources and empirical initiatives available to track them. Finally, Section 4.2.3 identifies the most important remaining measurement gaps.

4.2.1 Indicators Across the Framework

Indicators of the Direct Effects

The first set of indicators concerns how AI is used in production (Section 3.1). Provider usage data can show whether AI primarily augments workers or performs tasks end-to-end, and which occupations are most affected. Adoption data can reveal whether use spreads across education levels, occupations, industries, and firms of different sizes or remains concentrated among highly educated workers and large firms. Labor-market outcomes then capture the consequences, including changes in wages, employment, and hiring in the most exposed occupations, especially clerical and entry-level professional work, and within adopting firms. They can also show whether productivity gains accrue primarily to less-experienced workers or to those already at the top.

Worker-transition data indicate whether displaced workers move into similarly paid jobs or fall down the wage distribution. Job postings can reveal whether genuinely new occupations are emerging, how well they pay, how broadly accessible they are, and whether employers are raising or lowering skill requirements in existing jobs. Finally, the cost of using AI to perform tasks end-to-end relative to wages provides a direct measure of firms' incentives to automate rather than augment workers.

Indicators of Who Captures the Gains

The second set of indicators tracks how productivity gains are divided (Section 3.2). Upstream, the margins and market shares of AI suppliers can reveal how much of the gains are captured as rents, while the prices and capacity of key inputs—compute, advanced packaging, memory, and electricity—show how binding supply bottlenecks remain. Financial markets add a forward-looking but noisy signal: the valuations of AI suppliers reflect expected future profits, which combine expectations about market growth, margins, and the persistence of rents and bottlenecks.

Downstream, markups and concentration in AI-adopting industries can reveal how the remaining gains are split among firms, workers, and consumers: whether prices fall where AI cuts costs, and whether wages track productivity in the firms that adopt it. The responsiveness of demand in AI-exposed sectors can help determine whether productivity gains translate primarily into higher output and employment or into lower prices for consumers. Inflation measured separately for different consumption baskets can reveal whose cost of living benefits most from AI-driven price declines (Jaravel, 2019).

Indicators of Economy-Wide Demand Effects

The third set of indicators tracks how AI's gains spread through the economy (Section 3.3). Household saving and debt can show whether the gains are primarily spent, accumulated as wealth, or used to finance borrowing elsewhere. Individual-level consumer surveys can reveal how much AI's beneficiaries are likely to spend or save and which goods and services they are likely to purchase. Changes in the sectoral composition of employment can then show where the resulting demand lands and which workers ultimately benefit, including whether employment growth remains concentrated in professional services or extends to in-person and middle-wage occupations.

The Scoreboard

Ultimately, the clearest evidence comes from the three inequality channels themselves. Distributional national accounts track labor-income concentration (c_L), the capital share (s_K), and capital-income concentration (c_K) (Piketty et al., 2018). These are lagging indicators: they are measured relatively infrequently and may respond only gradually to technological change. Yet they provide the ultimate test of which distributional path is emerging.

Table 2: Indicators of Concentrated Gains and Shared Prosperity

Indicator	Signal of concentrated gains	Signal of shared prosperity
Direct effects on production (Section 3.1)		
Automation versus augmentation in AI use	Increasing end-to-end automation, especially in middle-paying and entry-level work	AI remains primarily assistive and raises workers' productivity
Adoption across workers and firms	Use remains concentrated among highly educated workers and large firms	Use spreads across education levels, occupations, industries, and firm sizes
Productivity gains across workers	Gains favor experienced, highly paid, or managerial workers	Gains are largest for less-experienced and lower-paid workers
Wages, employment, and hiring in exposed work	Falling wages and employment in entry-level and lower-skilled jobs	Rising employment and wages for lower-skilled and less-experienced workers
Worker transitions after displacement	Displaced workers move into lower-paid work or leave employment	Displaced workers move into similarly paid or better-paid work
New tasks, occupations, and skill requirements	Few new roles emerge, and access requires advanced technical or managerial skills	New roles emerge at scale and are accessible to a broad range of workers
Cost of end-to-end AI automation relative to wages	Falls below labor costs across a growing range of tasks	Remains high for reliable autonomous performance
Division of productivity gains (Section 3.2)		
AI suppliers' margins and market shares	Persistent high margins and concentration among a small number of suppliers	Entry, competition, and open models reduce margins and concentration
Prices and capacity of key AI inputs	Persistent shortages and scarcity rents in compute, memory, packaging, or energy	Capacity expands and input prices decline
Market valuations of AI suppliers	Valuations imply expectations of persistent margins, bottlenecks, and rents	Valuations imply expectations of stronger competition and declining margins
Profits, wages, and prices in adopting firms	Gains appear mainly as higher profits, with little wage or price pass-through	Gains are shared through higher wages and lower consumer prices
Wages relative to productivity in adopting firms	Productivity rises substantially faster than worker compensation	Worker compensation rises alongside productivity
Output and employment responses in AI-exposed sectors	Output responds weakly and employment declines	Output expands enough to sustain or increase employment
Price changes across household consumption baskets	Price declines favor goods and services consumed disproportionately by high-income households	Price declines extend to widely consumed goods and essential services
Economy-wide demand effects (Section 3.3)		
Household spending, saving, and debt	Saving and wealth rise at the top while borrowing rises below	Spending and asset accumulation increase broadly across households
Composition of household spending	Additional spending is concentrated in skill-intensive services	Spending expands across labor-intensive, in-person, and broadly consumed services
Sectoral composition of employment growth	Employment growth is concentrated in high-skill professional occupations	Growth extends to middle-wage and in-person occupations
Inequality outcomes (Section 2)		
Labor-income concentration (c_L)	Wage inequality rises	Wage inequality stabilizes or declines
Capital share (s_K)	Income shifts from labor toward capital	The labor share stabilizes or rises
Capital-income concentration (c_K)	Capital income becomes more concentrated	Capital income becomes less concentrated

Notes: The columns describe directional signals rather than numerical thresholds. No single indicator is decisive, and indicators may temporarily move in different directions. The strongest evidence will come from their joint movement across the three stages of the framework and the three inequality outcomes.

4.2.2 Existing Data and Measurement Initiatives

Monitoring these indicators would not begin from scratch. Official statistics provide the foundation. BLS data track employment, wages, and prices; household surveys such as the CPS and the Consumer Expenditure Survey underpin decades of research on wage inequality and consumption (Katz and Autor, 1999; Autor et al., 2020b); the Census Bureau's Business Trends and Outlook Survey provides frequent estimates of firm-level AI adoption (U.S. Census Bureau, 2026); and distributional national accounts measure the three inequality channels themselves (Piketty et al., 2018).

A complementary measurement ecosystem has also emerged around these sources. Companies such as Lightcast and Revelio Labs compile near-real-time data from job postings and online worker profiles. AI providers publish usage data through initiatives such as the Anthropic Economic Index, OpenAI's usage studies, and Google's ATLAS (Anthropic, 2025; Chatterji et al., 2025; Iscenko et al., 2026). Epoch AI collects systematic data on compute, hardware, and the AI supply chain (Somala, 2026), while dedicated surveys track AI adoption among workers and firms (Bick et al., 2024; Lane et al., 2023).

Efforts to combine these sources are beginning to provide more integrated views of AI's economic effects. The Stanford Digital Economy Lab's AI Economic Indicators bring together payroll, survey, and macroeconomic data in regularly updated dashboards. Its Canaries Dashboard, for example, uses ADP payroll records to track employment by age and AI exposure (Stanford Digital Economy Lab, 2026; Brynjolfsson et al., 2025a). Other work links job postings to résumé histories to study how firm-level AI adoption changes hiring across seniority levels (Hosseini and Lichtinger, 2026b). Together, these initiatives provide early models for the monitoring system envisioned in this section.

4.2.3 Remaining Measurement Gaps

Despite this progress, several central components of the framework remain poorly measured or are not measured in ways that can be directly linked to AI. These include the cost of automating specific tasks; the creation of genuinely new tasks; the distribution of productivity gains across workers; market power and rents along the AI supply chain; the extent to which wages and prices adjust to productivity gains in adopting firms; the responsiveness of demand in AI-exposed sectors; the ownership of AI-related capital;

household saving responses; and the composition of additional spending.

Even where relevant measures exist, the evidence remains fragmented. AI providers observe how the technology is used but not its broader economic consequences, while statistical agencies observe economic outcomes but have limited visibility into AI use. A growing number of studies link AI adoption or exposure to labor-market outcomes, which is an important step, but establishing causal effects remains difficult. These analyses are also rarely linked to measures of market power, supply-chain bottlenecks, capital ownership, household income, or sectoral price and income elasticities of demand.

Building datasets that connect the indicators discussed in this section would provide a clearer picture of how AI's gains are distributed. Developing this infrastructure will be essential for determining which distributional path the economy is heading toward.

5 Conclusion

Artificial intelligence could raise living standards broadly, or it could concentrate its gains at the top while leaving much of the workforce behind. How those gains are distributed will have profound economic, social, and political consequences for decades to come. This paper has developed a unified framework for understanding the forces that will shape that distribution.

The framework distinguishes three channels of inequality: the concentration of labor income (c_L), the division of income between labor and capital (s_K), and the concentration of capital income (c_K). It then traces how AI affects these channels through three stages: its direct effects on production, the initial division of productivity gains, and the economy-wide demand effects generated as those gains are spent or saved. The central conclusion is that AI does not imply a single distributional outcome. The same technology can produce concentrated gains or shared prosperity depending on what it can do, how firms deploy it, how markets divide the resulting gains, who owns the capital that benefits, and where subsequent demand falls. Some of these forces are technological, but many reflect market structure, bargaining power, access, and ownership.

The outcome is therefore not predetermined. Policies that shape competition, workers' bargaining power and mobility, access to AI and the skills needed to use it, and the ownership of AI-related capital can influence how gains are distributed before taxes and transfers. Redistribution can then offset some of the concentration that market outcomes still produce. Just as importantly, many of the mechanisms identified in this paper can be observed before they appear in aggregate inequality statistics. Monitoring AI use, labor-market outcomes, market power, the division of productivity gains, and the subsequent demand effects can provide early evidence of which path is emerging.

The distribution of AI's gains will be determined not by the technology alone, but by the markets, institutions, and policies surrounding it. Understanding those interactions is therefore essential to shaping an economy in which AI expands both prosperity and the number of people who share in it.

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A Appendix

A.1 Pre-Tax versus Post-Tax Income Inequality

This paper studies inequality in *pre-tax* income. The primary reason is conceptual: technology affects the income distribution through markets—wages, profits, and prices—so its direct effects are reflected primarily in the pre-tax distribution. Taxes and transfers affect how much of those effects ultimately reach households, but they are largely a policy response to technological change rather than part of its direct impact. Informing that policy response is one of this paper's objectives. This appendix establishes a second, empirical reason for focusing on pre-tax income: across countries, and over time within countries, post-redistribution income inequality closely tracks pre-redistribution inequality. In other words, historically, changes in pre-redistribution inequality have passed through to post-redistribution inequality almost one-for-one.

Data. We use the World Inequality Database (World Inequality Database, 2026), which builds on the distributional national accounts methodology (Piketty et al., 2018). For each economy and year, we take the shares of national income received by the top 1% and the bottom 50% under two income concepts: *pre-tax national income*, which includes social-insurance benefits net of contributions but excludes most other redistribution, and *post-tax national income*, measured after all taxes and transfers, including in-kind transfers. The sample covers 38 of the 43 economies in the IMF's April 2026 advanced-economy classification, excluding five whose WID series are entirely imputed from regional models.[16] The cross-sectional analysis uses 2022, a recent year with complete coverage, chosen to avoid reliance on the nowcast- and extrapolation-heavy 2024 endpoint; the panel specifications use an unbalanced panel of 1,777 country-years from 1913 to 2024.

Method. We regress each post-tax income share on the corresponding pre-tax share using four specifications: a 2022 cross-section, a pooled panel, a within-country specification with country fixed effects, and a specification with both country and year fixed effects. Two statistics are of primary interest. The R^2 measures how much of the variation in post-tax inequality is explained by pre-tax inequality. The slope measures the degree

[16] Andorra, Liechtenstein, Macao SAR, Puerto Rico, and San Marino, whose four series all carry a WID data-quality score of zero. Results including these economies are nearly identical. Note also that WID ranks the population separately under each income concept, so the pre-tax and post-tax top 1% need not contain exactly the same people.

of *pass-through*: how much of a cross-country difference or within-country change in pre-tax inequality carries over to post-tax inequality. In the figures, the dashed 45-degree line indicates one-for-one pass-through, while the vertical distance of the observations below that line reflects the level effect of redistribution.

Results. Figure A.1 presents the cross-country evidence for 2022. Redistribution reduces the level of inequality in every economy: the post-tax top-1% share lies below the corresponding pre-tax share in all 38 economies, while the post-tax bottom-50% share lies above it. Yet pre-tax inequality remains a strong empirical predictor of post-tax inequality. Across countries, it accounts for 80–85% of the variation in post-tax income shares, and differences in top income shares pass through almost one-for-one (slope = 0.93). For the bottom 50%, the estimated slope exceeds one (1.38): countries where the bottom half receives a larger pre-tax income share also redistribute more toward it, amplifying rather than offsetting cross-country differences. Figure A.2 presents the corresponding within-country evidence over time. Historically, changes in pre-tax inequality passed through to post-tax inequality at 0.83–0.88, with within-R^2 values of 0.74–0.89. Adding year fixed effects changes these estimates little (Table A.1).

Interpretation. Over the past century, post-tax inequality has tracked pre-tax inequality very closely. Taxes and transfers reduce the level of inequality, but they have not systematically offset differences in pre-tax inequality across countries or changes within countries over time. How AI shapes the pre-tax distribution is therefore likely to be the primary determinant of its ultimate effect on the distribution of household income. This motivates the paper's focus on pre-tax inequality.

Table A.1: Post-Tax on Pre-Tax Income-Share Regressions, IMF Advanced Economies

	Top 1% share		Bottom 50% share		
Specification	Slope	R^2	Slope	R^2	Obs.
Cross-section (2022)	0.925 (0.085)	0.849	1.383 (0.097)	0.800	38
Pooled panel	0.866 (0.031)	0.882	1.225 (0.083)	0.742	1,777
Country fixed effects	0.830 (0.031)	0.892	0.881 (0.047)	0.737	1,777
Country and year fixed effects	0.896 (0.043)	0.867	1.073 (0.051)	0.785	1,777

Notes: Each cell reports the coefficient from a regression of the post-tax income share on the corresponding pre-tax income share. Standard errors in parentheses: clustered by economy in the panel specifications. In the fixed-effects rows, R^2 is the within R^2.

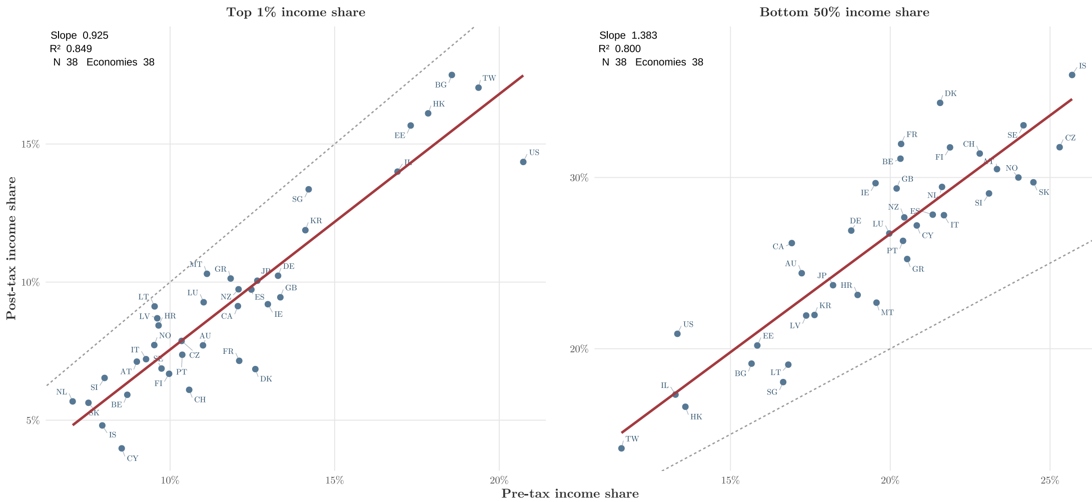

Figure A.1: Pre-Tax versus Post-Tax Income Shares Across Advanced Economies, 2022

Notes: Each point is one of 38 IMF advanced economies in 2022. The dashed 45-degree line marks one-for-one pass-through: points below it (top 1%) or above it (bottom 50%) indicate that redistribution reduces inequality in levels. Slopes are estimated with heteroskedasticity-robust standard errors. Source: World Inequality Database.

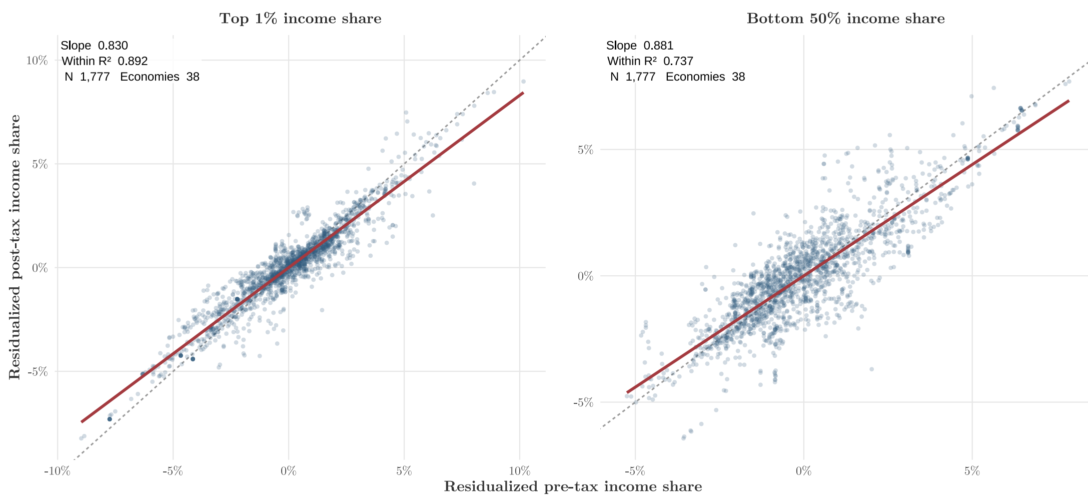

Figure A.2: Pre-Tax versus Post-Tax Income Shares Within Countries over Time

Notes: Both income shares are residualized on country fixed effects, so each point represents a country-year deviation from the country's own average (1,777 country-years, 38 economies, 1913–2024). By the Frisch–Waugh–Lovell theorem, the plotted slope equals the country fixed-effects estimate; the reported R^2 is the within R^2. The dashed 45-degree line marks one-for-one pass-through of movements. Standard errors are clustered by economy. Source: World Inequality Database.

A.2 Derivation of the Top-Income-Share Decomposition

Let Y denote total income, Y_{top} the income received by the top group, and L_{top} and K_{top} that group's labor and capital income. Then:

$$\begin{aligned}
S_{\text{top}} \equiv \frac{Y_{\text{top}}}{Y} &= \frac{L_{\text{top}} + K_{\text{top}}}{Y} \\
&= \frac{\frac{L_{\text{top}}}{L_{\text{total}}} L_{\text{total}} + \frac{K_{\text{top}}}{K_{\text{total}}} K_{\text{total}}}{Y} \\
&= c_L \frac{L_{\text{total}}}{Y} + c_K \frac{K_{\text{total}}}{Y} \\
&= (1 - s_K) c_L + s_K c_K.
\end{aligned}$$

Here, $c_L = L_{\text{top}}/L_{\text{total}}$, $c_K = K_{\text{top}}/K_{\text{total}}$, and $s_K = K_{\text{total}}/Y$.

A.3 An Analogous Decomposition for the Gini Coefficient

The decomposition in the main text is exact because the top income share is linear in incomes (it is a share of a sum). The most familiar inequality measure, the Gini, admits an analogous decomposition by income source, at the cost of one extra term. Writing total income as labor plus capital income, the Gini of total income decomposes as

$$G \;=\; \underbrace{(1 - s_K)}_{\text{labor share}} \; G_L \, R_L \;+\; \underbrace{s_K}_{\text{capital share}} \; G_K \, R_K,$$

where G_L, G_K are the Ginis of labor and of capital income (each source's own concentration) and R_L, R_K are the *Gini correlations* between each source and total income (how each source's ranking lines up with the ranking of total income).[17] This mirrors the three factors of the main text: each source enters through its **income share** (s_K, $1 - s_K$) and its **own concentration** (G_K, G_L). The difference is the correlation term R_k: with the Gini the labor–capital overlap appears as an explicit factor, whereas in the top-share formula it is absorbed by defining the top group by total income. The economic logic is the same—a rising capital share raises overall inequality more when capital income is both unequally

[17] $R_k = \operatorname{Cov}\big(y_k, F(y)\big) / \operatorname{Cov}\big(y_k, F(y_k)\big)$, where $F(y)$ is the rank by total income and $F(y_k)$ the rank by source k; $R_k \in [-1, 1]$.

distributed (G_K high) and concentrated among the rich (R_K high)—but the expression is no longer a simple weighted average.

A.4 Data and construction of Figure 2

Figure 2 draws on task-level automatability ratings for U.S. occupations from Hosseini and Lichtinger (2026a), which build on the task-based exposure framework of Eloundou et al. (2024). Eloundou et al. rate each task in the O*NET database for its exposure to large language models and, in an exploratory rubric, for its automation potential on a five-level scale—from tasks an AI system cannot reliably perform at all, to tasks it could complete on its own without human oversight. Hosseini and Lichtinger re-rate tasks for automatability with a more capable model, assigning each of roughly 19,000 tasks across 923 occupations an automation tier from T0 (least automatable) to T4 (most). These ratings describe what AI could *potentially* automate, not what has been automated: they are produced by language models applying a rubric, and the automation rubric in particular is exploratory rather than validated against realized outcomes.

The figure treats this automation tier as a threshold and reports three scenarios: *limited* counts only fully-automatable tasks (T4); *modest* counts tasks rated clearly automatable (T3–T4, the binary measure used in the main text); and *broad* counts any task with at least partial automation potential (T2–T4). For each occupation we compute the share of its tasks automatable under each threshold, weighting tasks by their share of the occupation's labor time. We then attach each occupation's median annual wage and employment from the BLS Occupational Employment and Wage Statistics (2024), rank occupations by wage, and group them into employment-weighted deciles, each holding about a tenth of U.S. workers. The figure plots the employment-weighted average automatable share within each decile, separately for the three scenarios (864 occupations, about 142 million workers; because employment is reported at a broader occupational level, it is split evenly across the detailed occupations within each group to avoid double-counting). The underlying data and code are available from the authors upon request.